

Hanyeol Ryu

Research Assistant

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RESEARCH INTERESTS

- Low-power wireless communications (e.g. backscatter communications)
- Channel modeling & estimation
- AI/ML for intelligent wireless communications
- Wireless security & privacy
- Software-defined radio

EDUCATION

Pusan National University

- *B.S. in Electronics Engineering* Feb 2026

RESEARCH EXPERIENCE

Pusan National University

- *Undergraduate Researcher (PI: Prof. Sangkil Kim)* Jan 2024 – Feb 2026
- *Research Assistant (PI: Prof. Sangkil Kim)* Mar 2026 – Present

PUBLICATIONS

Published Articles (†: First author, *: Corresponding author)

- **J4:** H. Ryu[†], N. Ha, and S. Kim*, “Transformer-Hypernetwork-Controlled Deep-Unfolded Phase-Aware Channel Estimation Refinement for Phase-Drift-Robust Backscatter Links,” *arXiv preprint*, Jun. 2026. (DOI: [10.48850/arXiv.2606.31400](https://doi.org/10.48850/arXiv.2606.31400)), (*Wireless Communications Letters*, Under Review).
- **J3:** N. Ha[†], H. Ryu, H. Jeong, H. Kim, G. Kim, A. Georgiadis, M. M. Tentzeris, and S. Kim*, “Integrated Sensing and Backscatter Communication (ISABC): Toward Self-Sustainable Autonomous IoT Networks,” *IEEE Microwave Magazine*, Accepted, 2026.
- **J2:** H. Ryu[†], and S. Kim*, “Ultra-low-power Monostatic Backscatter Platform with Phase-Aware Channel Estimation and System-Level Validation,” *IEEE Internet of Things Journal*, May 2026. (DOI: [10.1109/JIOT.2026.3696951](https://doi.org/10.1109/JIOT.2026.3696951))
- **J1:** J. Choi[†], H. Ryu[†], and S. Kim*, “Improved Backscattering Communication Range Using Reflection Amplifiers,” *The Journal of Korean Institute of Electromagnetic Engineering and Science*, Feb. 2025. (DOI: [10.5515/KJKIEES.2025.36.3.266](https://doi.org/10.5515/KJKIEES.2025.36.3.266))
- **C2:** H. Ryu[†], and S. Kim*, “Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links,” *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Accepted, 2026.
- **C1:** H. Ryu[†], and S. Kim*, “High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System,” *IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI)*, 2025. (DOI: [10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818](https://doi.org/10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818))

PATENTS

- Korean Patent: “Apparatus and Method for Phase-Aware Channel Estimation in Monostatic Backscatter Communication Systems”, Apr. 2026.

CONFERENCE PRESENTATIONS

Title: Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links (Oral)

- *IEEE ISPACS*

Oct 2026

Title: Ultra-low-power Backscatter Platform with Phase-Aware Channel Estimation (Oral)

- *The Korean Institute of Electromagnetic Engineering and Science (KIEES)*

Feb 2026

Title: High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System (Oral)

- *IEEE AP-S/URSI*

Jul 2025

Title: Improved Backscattering Communication Range Using Reflection Amplifiers (Poster)

- *KIEES*

Feb 2025

PROFESSIONAL ACTIVITIES

- TPC Member, *IEEE ISPACS*, 2026.

HONORS & AWARDS

- 2nd Place: Hackathon, **Pusan National University**, Aug. 2023.
- 2nd Place: Capstone Design Project, **Pusan National University**, Nov. 2025.

SERVICE & EXTRACURRICULAR ACTIVITIES

University

Pusan National University

- *Treasurer, Student Council* Dec 2021 – Dec 2022
- *Student Ambassador, Pusan National University* Apr 2021 – Feb 2023
- *Volunteer, Overseas Volunteering Program (3D Printer & Arduino) — Thailand* Dec 2022 – Feb 2023

Military Service

Republic of Korea Marine Corps (ROKMC)

- *Squad Leader, ROKMC Special Reconnaissance* Feb 2017 – Jul 2018
- 2nd Place, *ROKMC Special Reconnaissance Selection and Qualification Training* Jun 2017 – Aug 2017

SKILLS & TECHNIQUES

- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **SDR Platform:** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7